Light

GCSE Computer Science — Memory and Storage (Set 2)

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GCSE • OCR J277 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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← Back to Memory And Storage

Multiple-choice topic test for **GCSE Computer Science (OCR J277)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

When you open a word processor application, where is the program loaded to?

- **A.** RAM
- **B.** ROM
- **C.** The CPU cache
- **D.** The optical drive

Q2 (1 mark)

Which of the following is an example of data stored in ROM?

- **A.** A user's holiday photographs

- ○ **B.** The firmware that starts the computer when powered on
- ○ **C.** A document currently being edited
- ○ **D.** Music files downloaded from the internet

Q3 (1 mark)

Where is virtual memory physically located?

- ○ **A.** On special memory chips inside the CPU
- ○ **B.** On a removable USB drive
- ○ **C.** On the secondary storage device (hard drive or SSD)
- ○ **D.** In the cloud via an internet connection

Q4 (1 mark)

What material are the platters in a hard disk drive coated with?

- ○ **A.** A magnetic material that can be magnetised in different directions
- ○ **B.** A reflective silver coating
- ○ **C.** Silicon semiconductor material
- ○ **D.** Optical glass for laser reading

Q5 (1 mark)

What type of memory technology do Solid State Drives use to store data?

- ○ **A.** Magnetic storage on spinning discs
- ○ **B.** Flash memory using transistors that trap electrical charges
- ○ **C.** Optical pits burned by a laser
- ○ **D.** Volatile memory that requires constant power

Q6 (1 mark)

What does the 'RW' stand for in CD-RW?

- ○ **A.** Read Widely
- ○ **B.** Random Write
- ○ **C.** ReWritable
- ○ **D.** Rapidワークス

Q7 (1 mark)

How many different values can a single bit represent?

- ○ **A.** 2 values
- ○ **B.** 1 value
- ○ **C.** 8 values

- ○ **D.** 256 values

Q8 (1 mark)

Which is the largest unit of storage?

- ○ **A.** Gigabyte (GB)
- ○ **B.** Megabyte (MB)
- ○ **C.** Kilobyte (KB)
- ○ **D.** Terabyte (TB)

Q9 (1 mark)

Why is RAM described as 'volatile' memory?

- ○ **A.** Because it is dangerous and can explode
- ○ **B.** Because it loses all stored data when power is removed
- ○ **C.** Because data can be written very quickly
- ○ **D.** Because its contents change frequently during use

Q10 (1 mark)

Rank these storage types from fastest to slowest access speed.

- ○ **A.** HDD → SSD → RAM

- ○ **B.** RAM → HDD → SSD
- ○ **C.** RAM → SSD → HDD
- ○ **D.** SSD → RAM → HDD

Q11 (1 mark)

Why do many budget laptops still include HDDs rather than SSDs?

- ○ **A.** HDDs offer more storage capacity for less money
- ○ **B.** HDDs are faster than SSDs
- ○ **C.** HDDs are more durable than SSDs
- ○ **D.** HDDs use less power than SSDs

Q12 (1 mark)

What does 'colour depth' measure in a digital image?

- ○ **A.** The physical size of the image when printed
- ○ **B.** The number of bits used to represent the colour of each pixel
- ○ **C.** How many pixels the image contains
- ○ **D.** The file format of the image

Q13 (2 marks)

Compare RAM and ROM in terms of their volatility and typical use.

- ○ **A.** Both are volatile and store user files
- ○ **B.** Both are non-volatile and store system instructions
- ○ **C.** RAM is non-volatile and stores firmware; ROM is volatile and stores running programs
- ○ **D.** RAM is volatile and stores running programs; ROM is non-volatile and stores firmware

Q14 (2 marks)

What happens during the 'paging' process in virtual memory management?

- ○ **A.** Data is moved between RAM and secondary storage in fixed-size blocks called pages
- ○ **B.** The computer displays pages of a document on screen
- ○ **C.** The printer outputs pages of text
- ○ **D.** Web pages are downloaded from the internet

Q15 (2 marks)

A folder contains 25 files, each exactly 40 MB in size.
What is the total size of all files in gigabytes (GB)?

- ○ **A.** 0.1 GB
- ○ **B.** 10 GB
- ○ **C.** 1 GB

○ **D.** 100 GB

Q16 (2 marks)

Over time, an HDD becomes 'fragmented' and performance decreases. What causes fragmentation and why does it affect performance?

○ **A.** The magnetic coating wears off the platters

○ **B.** Files become split into pieces stored in non-contiguous locations, requiring the read/write head to move more to access complete files

○ **C.** The spinning motor slows down over time

○ **D.** Dust particles block the laser from reading data

Q17 (2 marks)

SSDs have a limited number of write cycles per cell.
What does this limitation mean for users?

○ **A.** SSDs can only store a certain amount of data before becoming full

○ **B.** SSDs become faster as more data is written to them

○ **C.** Users must manually refresh the data on SSDs every month

○ **D.** After many write operations, cells may wear out and fail, though modern SSDs have techniques to distribute wear and extend lifespan

Q18 (2 marks)

A dual-layer Blu-ray disc holds 50 GB, while a single-layer disc holds 25 GB.
How does the dual-layer technology achieve this doubled capacity?

- ○ **A.** By having two separate data layers that the laser can focus on at different depths
- ○ **B.** By using a laser twice as powerful
- ○ **C.** By spinning the disc twice as fast
- ○ **D.** By compressing all data to half its original size

Q19 (2 marks)

Unicode can use up to 32 bits per character, compared to ASCII's 7 bits. What advantage does this provide?

- ○ **A.** Unicode characters are displayed in more colours
- ○ **B.** Unicode files are smaller than ASCII files
- ○ **C.** Unicode can represent characters from virtually all world languages, including Chinese, Arabic, and emoji, while ASCII is limited to basic English characters
- ○ **D.** Unicode makes text processing faster

Q20 (2 marks)

A camera takes photos at 4000 × 3000 pixel resolution.
How many megapixels is this camera?

- ○ **A.** 7 megapixels
- ○ **B.** 14 megapixels

- ○ **C.** 120 megapixels
- ○ **D.** 12 megapixels

Q21 (2 marks)

A mono audio file is converted to stereo format.
How does this affect the file size?

- ○ **A.** File size remains the same
- ○ **B.** File size approximately doubles because two channels of audio data are stored instead of one
- ○ **C.** File size is halved because stereo compression is more efficient
- ○ **D.** File size increases by 10% for the additional channel header

Q22 (2 marks)

A server uses multiple hard drives configured to automatically duplicate data across drives.
What characteristic is being prioritised?

- ○ **A.** Access speed only
- ○ **B.** Storage capacity only
- ○ **C.** Reliability — if one drive fails, data is preserved on the other drives
- ○ **D.** Portability of the storage system

Q23 (2 marks)

An audio recording uses 8-bit sampling.
How many different amplitude levels can be represented?

- ○ **A.** 8 levels
- ○ **B.** 16 levels
- ○ **C.** 65,536 levels
- ○ **D.** 256 levels

Q24 (3 marks)

An uncompressed bitmap image is 800 × 600 pixels with 24-bit colour depth. The file is 1.44 MB.
After compression, the file size is 180 KB.
Calculate the compression ratio (original size ÷ compressed size).

- ○ **A.** 8:1
- ○ **B.** 1:8
- ○ **C.** 1:80
- ○ **D.** 80:1

Q25 (3 marks)

A computer has 8 GB of RAM and a traditional HDD for virtual memory. Another identical computer has 8 GB RAM but uses an SSD for virtual memory.
Analyse how their performance would compare when RAM is exhausted.

- ○ **A.** Both would perform identically since they have the same amount of RAM
- ○ **B.** The HDD system would perform better because HDDs have

larger capacity

○ **C.** The SSD system would perform significantly better because SSD read/write speeds are much faster, reducing the performance penalty when swapping data between RAM and virtual memory

○ **D.** Neither would use virtual memory because they both have 8 GB of RAM

Q26 (3 marks)

An embedded system in a car's engine management unit needs storage for its control program.
Analyse which storage type would be most suitable and explain why.

○ **A.** An HDD because it offers large capacity for extensive data logging

○ **B.** Flash memory/ROM because it's non-volatile, resistant to vibration and temperature extremes, has fast access, and retains data without power

○ **C.** An optical disc because it's cheap and widely available

○ **D.** RAM only because it provides the fastest access speed

Q27 (3 marks)

A podcast episode is recorded in mono at 22,050 Hz sample rate with 16-bit depth for 36 minutes.
Calculate the uncompressed file size in MB.

○ **A.** Approximately 47.6 MB

○ **B.** Approximately 190.5 MB

○ **C.** Approximately 381 MB

○ **D.** Approximately 95.3 MB

Q28 (3 marks)

Computers need to store both positive and negative numbers.
Analyse how using 'signed' binary representation affects the range of values that can be stored in a byte.

○ **A.** A signed byte uses one bit to indicate positive/negative, leaving 7 bits for the value, giving a range of -128 to 127 instead of 0 to 255 for unsigned

○ **B.** A signed byte can store 0 to 255; an unsigned byte can store -128 to 127

○ **C.** Signed and unsigned bytes can store exactly the same range of values

○ **D.** Signed bytes use two bytes of storage instead of one

Q29 (3 marks)

A business is deciding between local storage (NAS device) and cloud storage for their files.
Analyse the key trade-offs between these approaches.

○ **A.** Cloud storage is always faster and cheaper than local storage

○ **B.** Local storage is completely secure while cloud storage has no security

○ **C.** Local NAS provides faster access on-site and no ongoing fees but requires maintenance and has limited off-site protection; cloud offers anywhere-access and off-site backup but has recurring costs and depends on internet connectivity

○ **D.** There are no significant differences — both provide identical

functionality

Q30 (3 marks)

A hard drive manufacturer advertises a drive as '2 TB capacity'. When installed, the operating system reports it as approximately 1.82 TB. Analyse why this discrepancy exists.

- ○ **A.** The manufacturer is lying about the capacity
- ○ **B.** Approximately 10% of the drive is defective
- ○ **C.** The operating system reserves 10% for virtual memory
- ○ **D.** Manufacturers use decimal units (1 TB = 1,000,000,000,000 bytes) while operating systems often use binary units (1 TB = 1,099,511,627,776 bytes), causing the apparent difference when the same bytes are measured differently

Total: **55** marks • Time: **30 min**

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